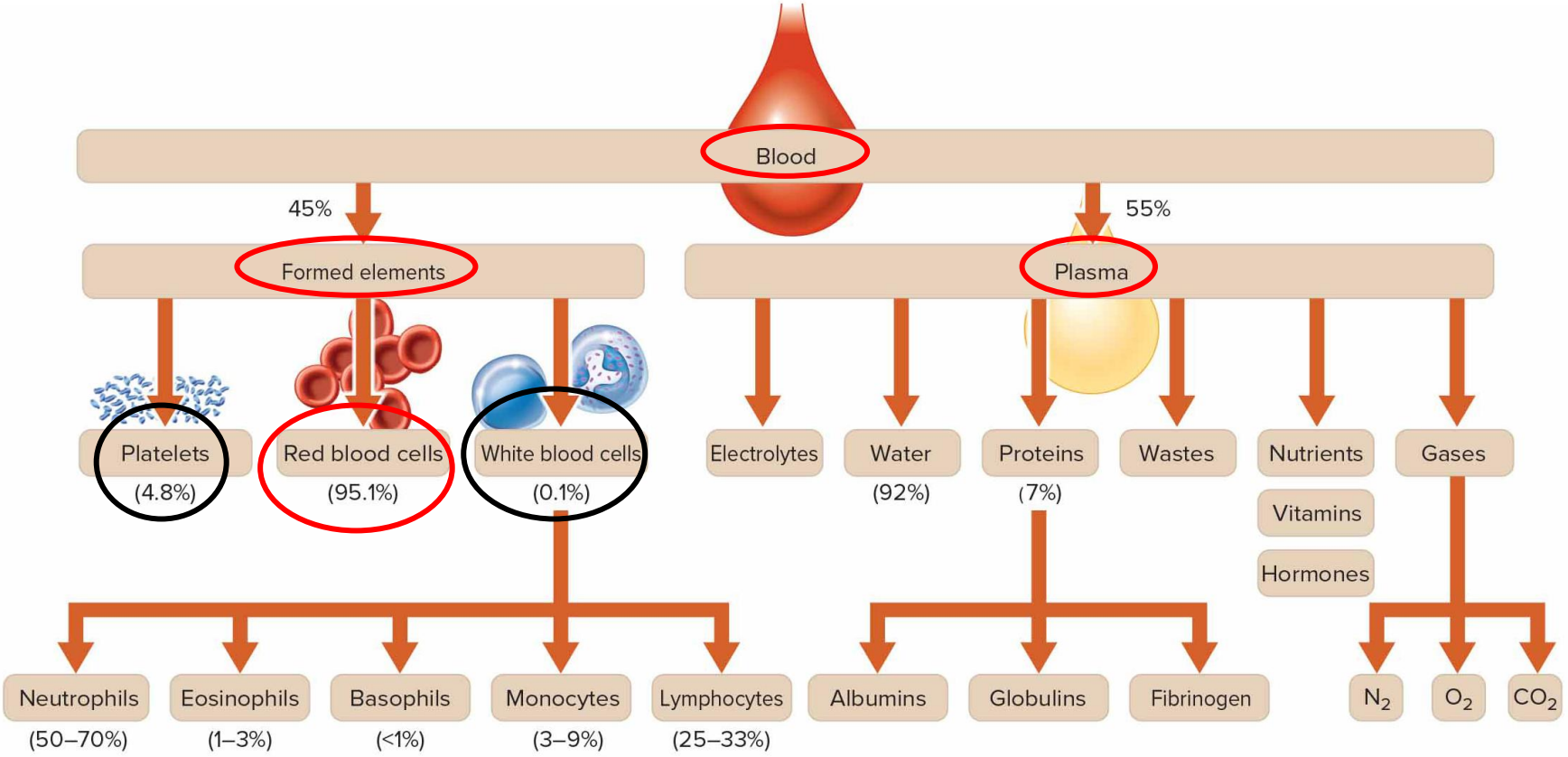


Anemia: Overview, Causes, and Management

Agenda

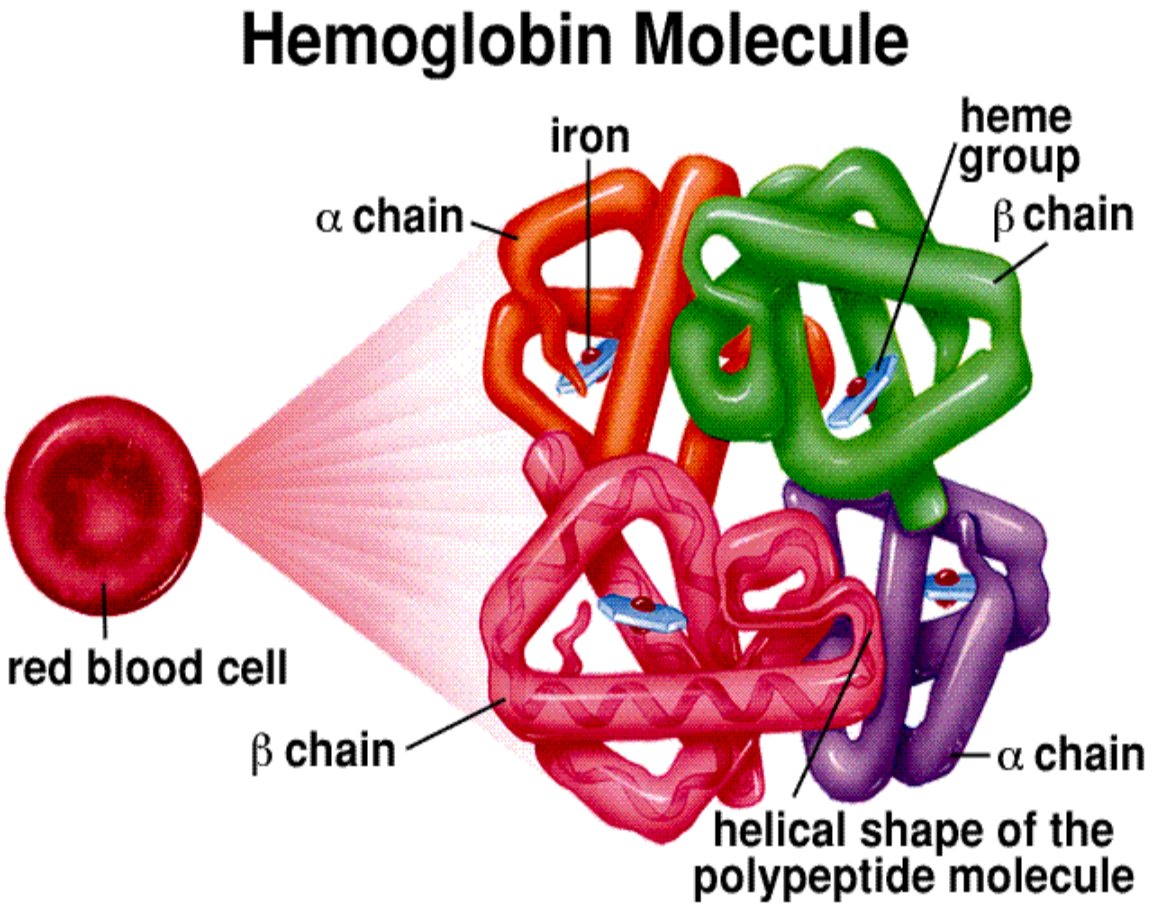
- Introduction to blood physiology
- Blood transfusion
- Anemias

BLOOD COMPOSITION: THE BIG PICTURE!

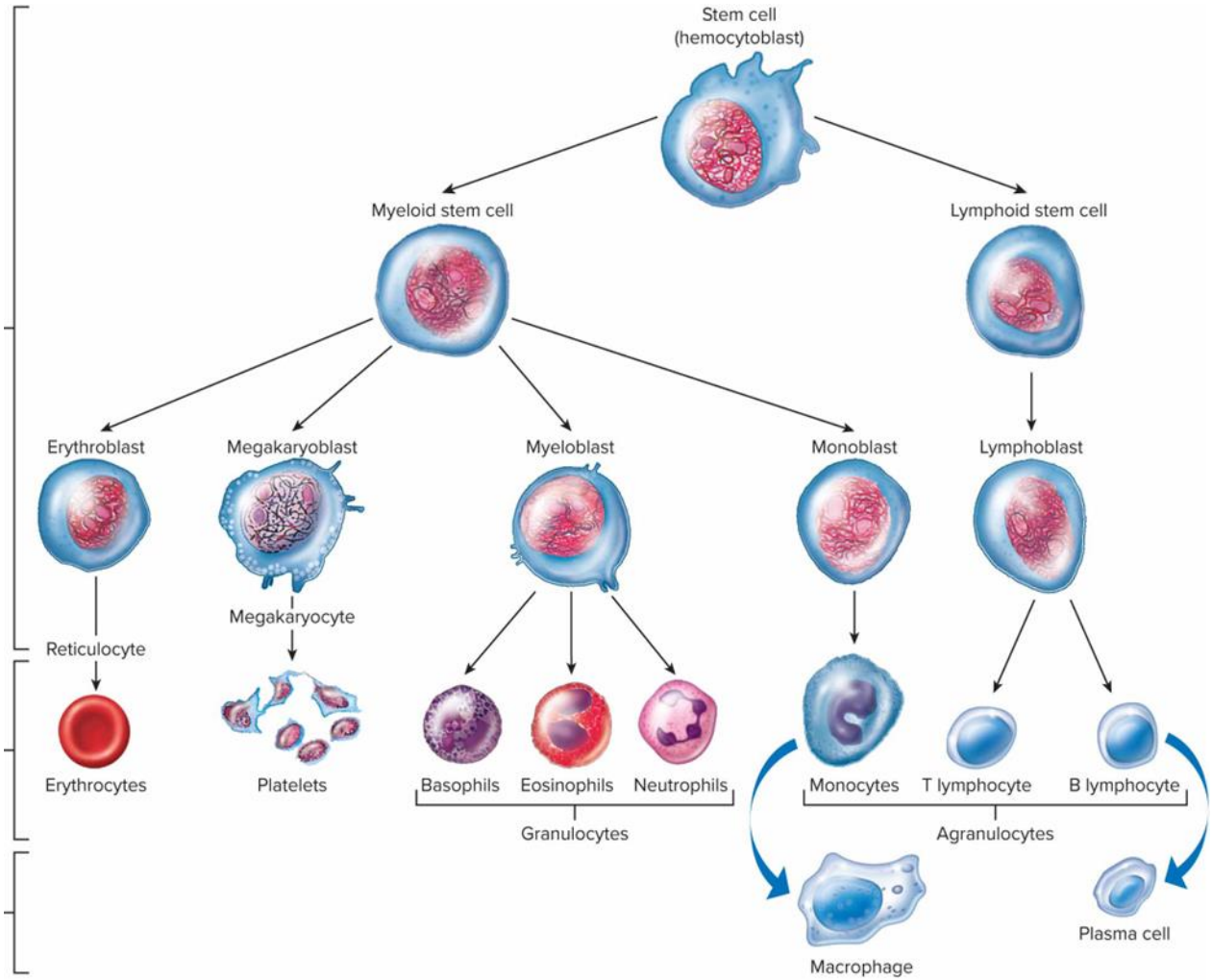


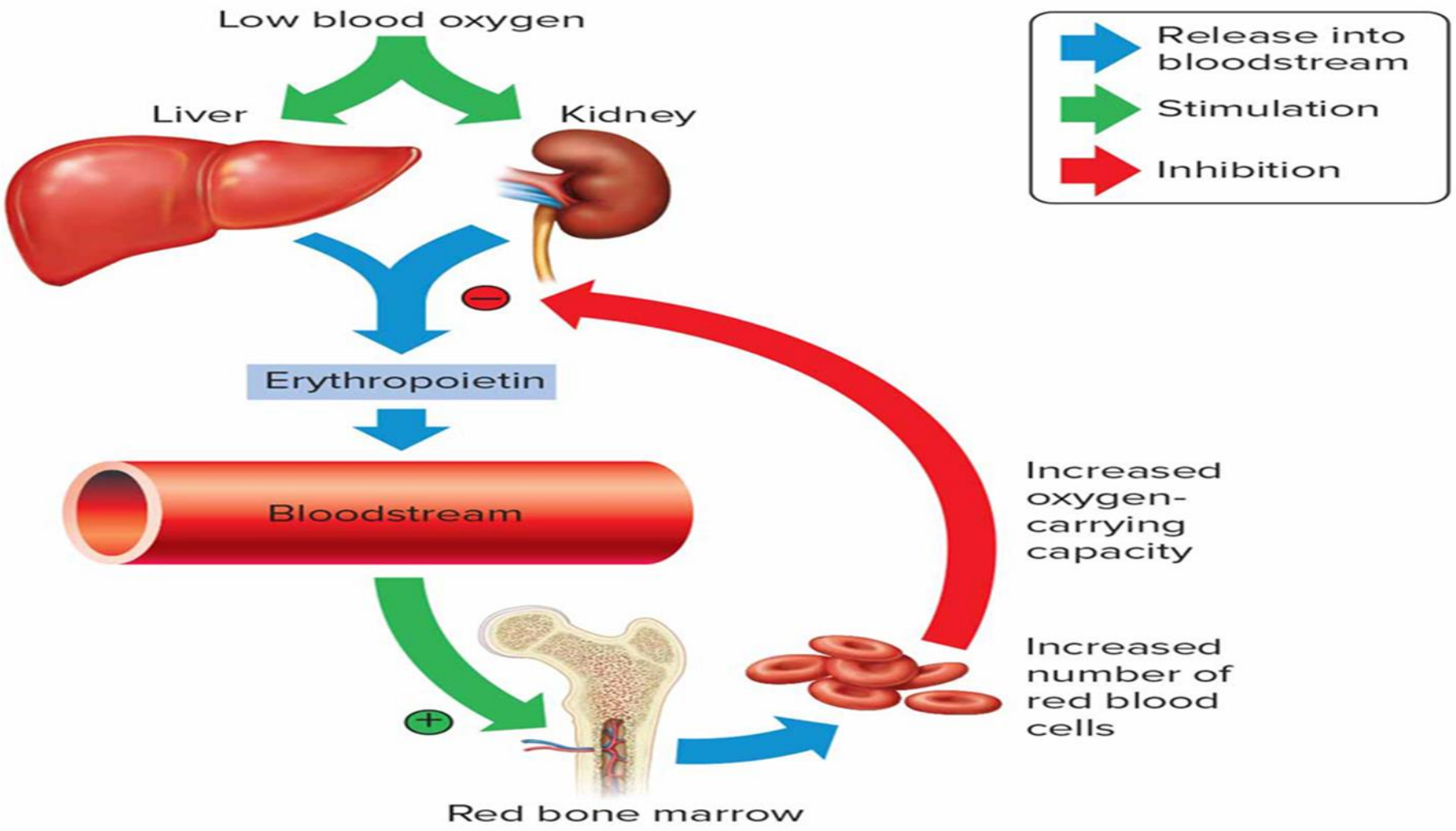
HEMOGLOBIN

- 4 folded globin, heme, & iron. 300 millions
- Adult HbA, HbA2, fetal HBF



ERYTHROPOIESIS




Low blood oxygen
Liver
Kidney
Release into bloodstream
Stimulation
Inhibition
Erythropoietin
Bloodstream
Increased oxygen-carrying capacity
Increased number of red blood cells
Red bone marrow

TRANSFUSION THERAPY

- Provides the means for replacement of red blood cells and other blood components
- Four major ABO blood types are determined by the presence or absence of two red cell antigens: A and B.
  - The presence of D antigen determines the Rh-positive type.
  - The absence of the D antigen determines the Rh-negative type.
  - https://youtu.be/cKnEdvrmHK4

RED BLOOD CELL COMPONENTS USED IN TRANSFUSION THERAPY

- Whole blood
- Red blood cells
- Leukocyte-reduced blood cells
- Washed red blood cells
- Frozen red blood cells

SIGNS AND SYMPTOMS OF TRANSFUSION REACTIONS

- Sensation of heat along the vein where the blood is being infused
- Flushing of the face
- Urticaria, headache, pain in the lumbar area
- Chills, fever, constricting pain in the chest
- Cramping pain in the abdomen
- Nausea, vomiting
- Tachycardia, hypotension, and dyspnea
- https://youtu.be/58Gp8Tiui1E

Definition of Anemia

- Reduction in RBC count, hemoglobin, or hematocrit
- Impact on oxygen delivery

TAXONOMY & CLINICAL PRESENTATION

kinetic approach: production, destruction, and removal

I. Faulty or decreased RBC production. (Iron Deficiency).
II. Excessive destruction of RBCs. (Sickle cell anemia).
III. Blood loss. (Hemorrhage).

Morphological approach: size, shape, number of RBCs, and the number of hemoglobin

1. Normocytic
2. Microcytic
3. Macrocytic

- https://youtu.be/NW30MvZVXoQ?si=dqoTV8PA7PU3LcmP

Causes of Anemia

- Decreased production: Deficiencies, bone marrow failure
- Increased destruction: Hemolysis, inherited disorders
- Blood loss: Acute (trauma) vs. chronic (GI bleeding)

Clinical Manifestations of Anemia

- Fatigue, pallor, dizziness, shortness of breath
- Severe cases: Tachycardia, heart failure, cognitive impairment

Iron-Deficiency Anemia

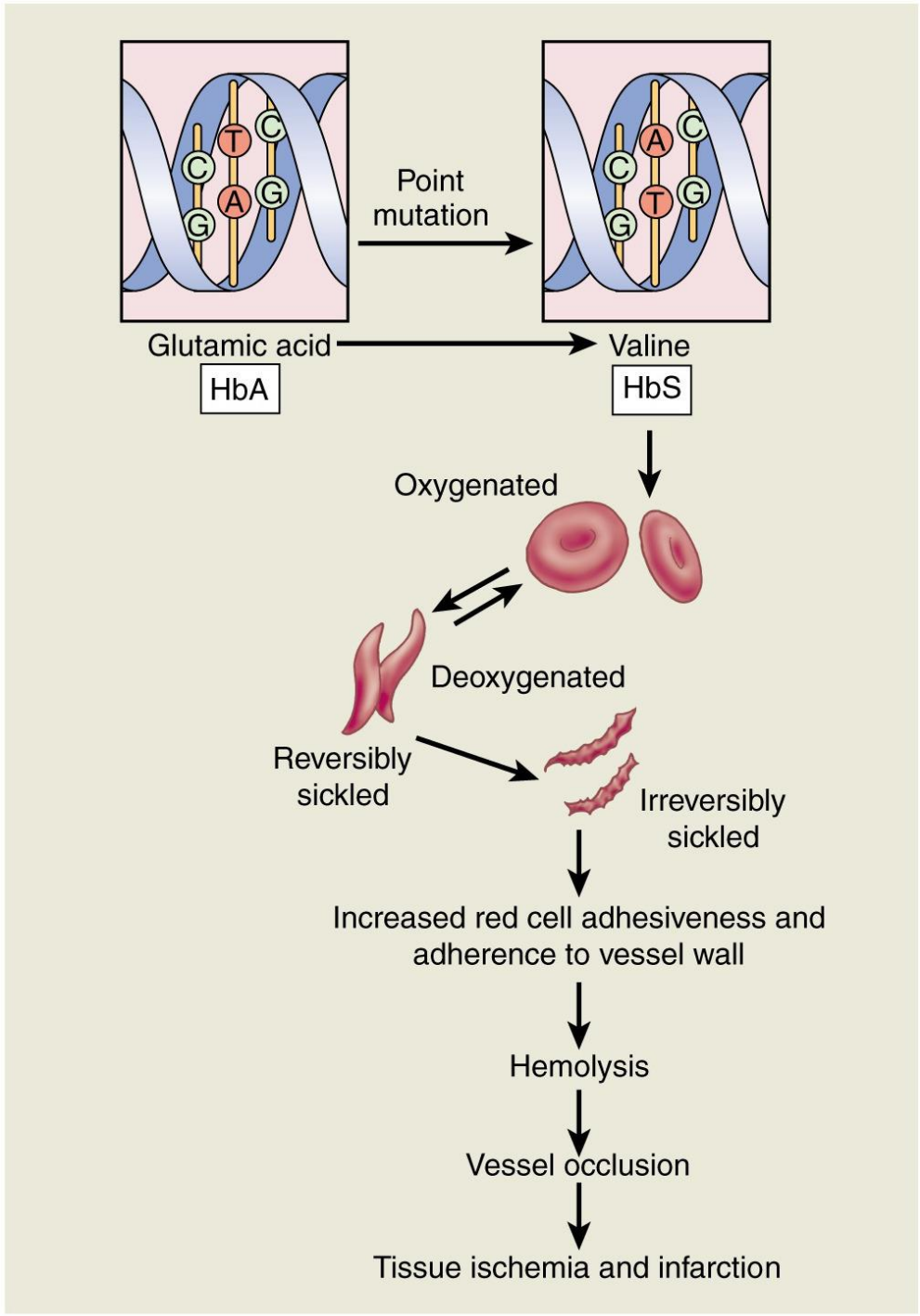
- Most common anemia type
- Causes: Poor diet, chronic blood loss, malabsorption
- Treatment: Iron supplementation

Vitamin B12, Folic Acid Deficiency Anemia

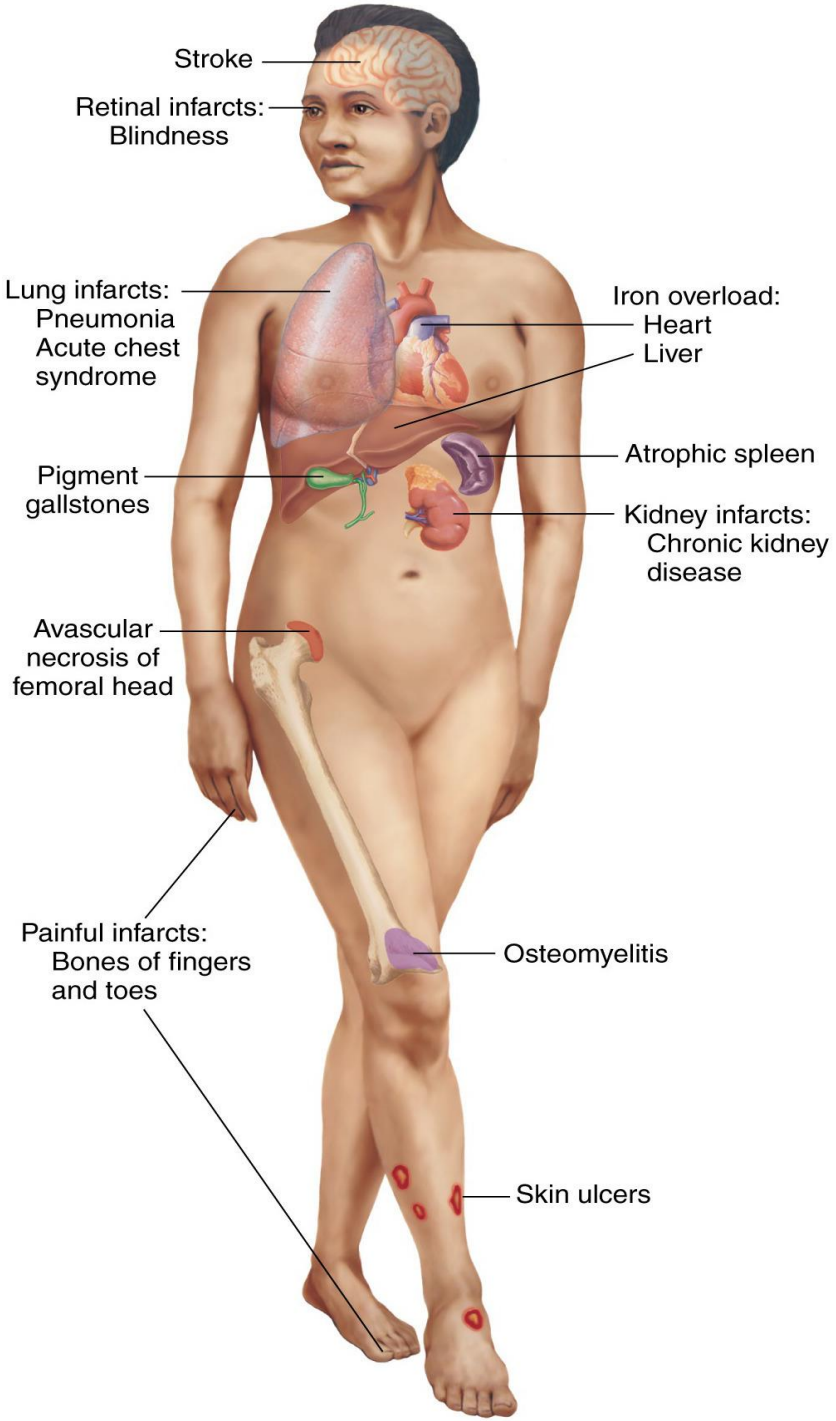
- Essential for DNA synthesis
- Causes: Pernicious anemia, malabsorption
- Symptoms: Macrocytic anemia, neurological deficits

Sickle Cell Disease

- Genetic mutation affecting HbS
- Causes vaso-occlusion, pain crises, organ damage
- Managed with hydroxyurea, transfusion, pain control



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Thalassemias

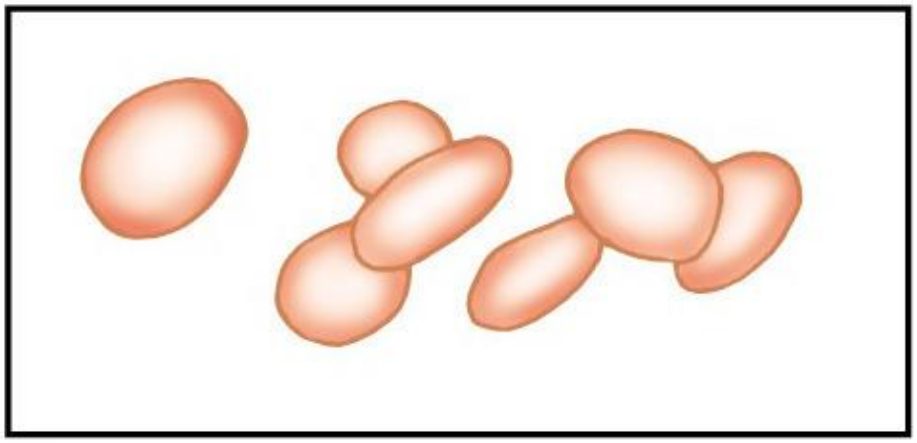
- Defects in alpha/beta globin chains
- Alpha-thalassemia, Beta-thalassemia major/minor
- Treatment: Transfusions, iron chelation, gene therapy

G6PD Deficiency

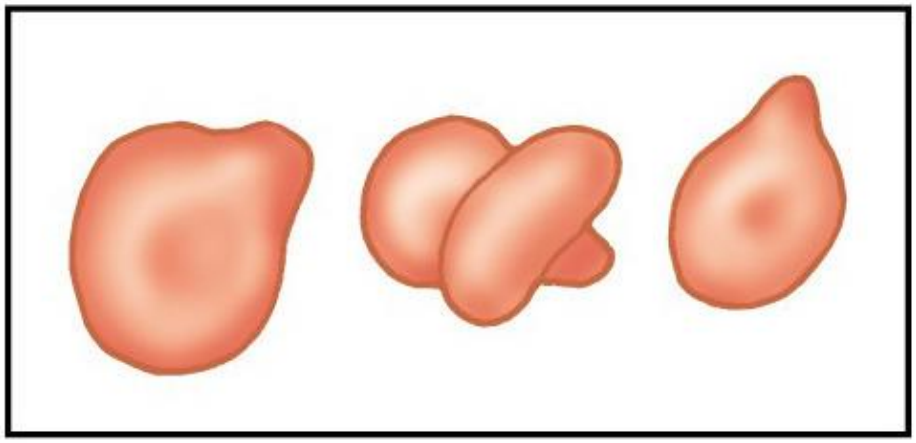
- X-linked disorder leading to hemolysis
- Triggered by infections, drugs, fava beans

Aplastic Anemia

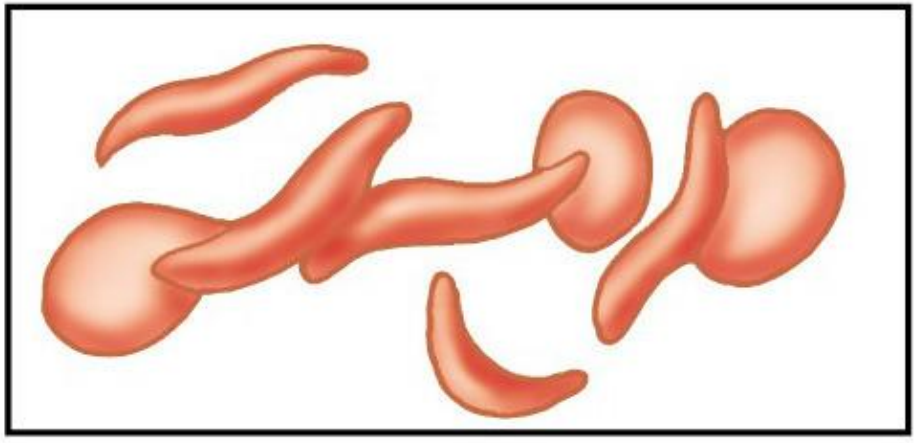
- Bone marrow failure, pancytopenia
- Causes: Autoimmune, drugs, radiation, infections
- Treatment: Immunosuppressants, bone marrow transplant



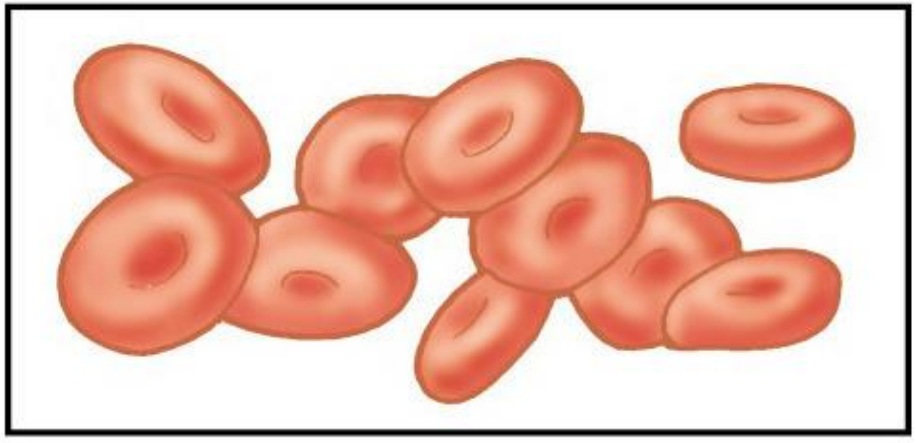
A Iron deficiency anemia



B Megaloblastic anemia



C Sickle cell disease



D Normal

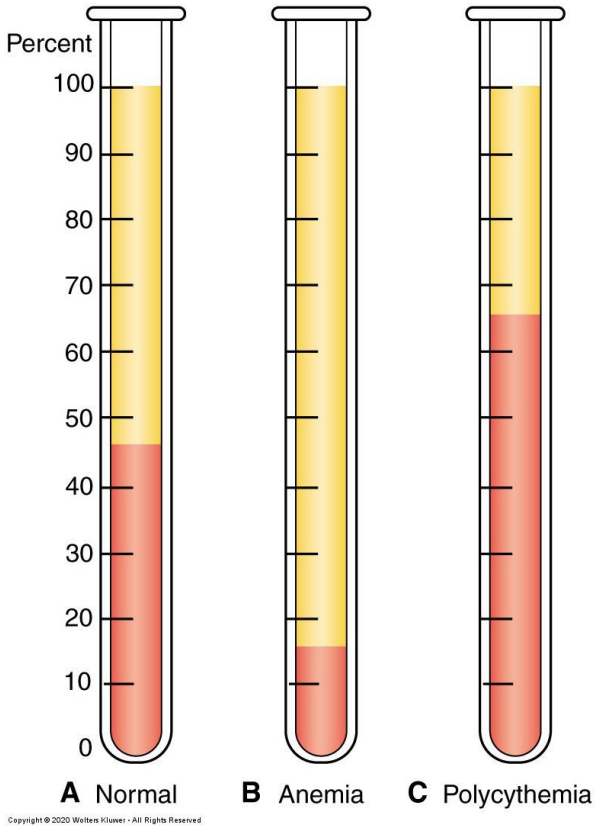
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Polycythemia

- Increased RBC mass
- Primary: JAK2 mutation
- Secondary: Hypoxia, EPO-producing tumors
- Management: Phlebotomy, hydroxyurea

Laboratory Tests for RBCs

- CBC: Hb, Hct, MCV, MCH, MCHC
- Peripheral blood smear
- Reticulocyte count, Iron studies, B12/folate levels



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Thank You